# PHARMATRUST: DECENTRALIZED PHARMACEUTICAL SUPPLY CHAIN PROVENANCE & ANTI-COUNTERFEIT PLATFORM

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## Abstract

Substandard and counterfeit pharmaceuticals claim over 1,000,000 lives annually a $200B+ in economic losses globally. Legacy supply chain architectures rely on cen databases that lack end-to-end transparency, enabling data manipulation, unautho duplication, and illicit diversion.

We present PharmaTrust, an industrial decentralized platform deployed on the Poly Layer-2 Proof-of-Stake blockchain. Batches are tokenized into ERC-721 non-fungib linked to IPFS clinical metadata, enforcing deterministic multi-hop role custody tran with zero-gas public verification for patients and pharmacies.

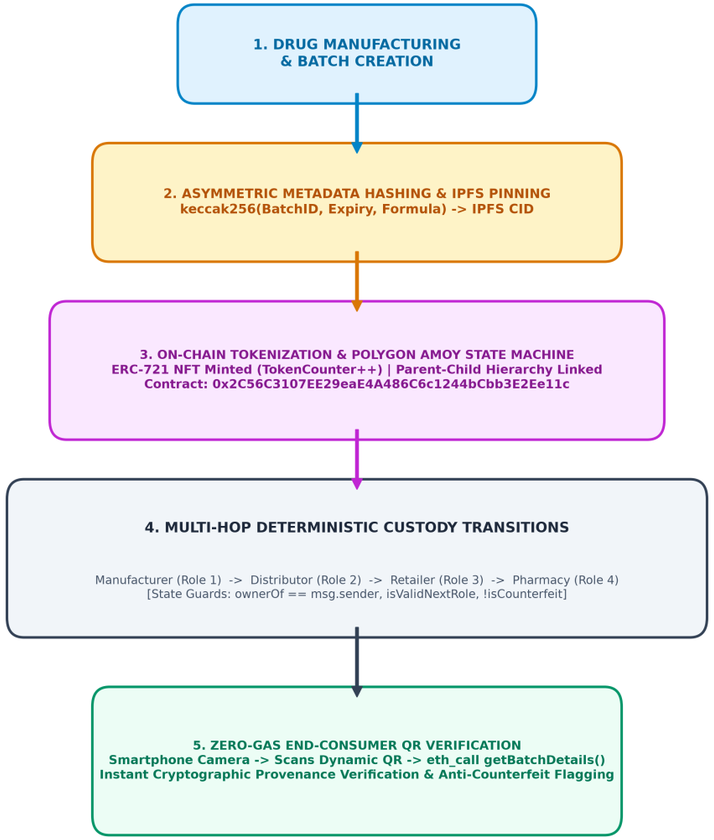
## Introduction : What Constitutes Security & Trust?

- **Batch Authenticity:** Cryptographic keccak256 fingerprinting of formula, dosage & permanently anchored on-chain with immutable timestamps.
- **Stakeholder Access Control:** Smart contract modifiers restricting custody handoffs strictly to verified regulatory roles (Mfg -> Dist -> Ret -> Pharm).
- **Zero-Gas Public Verification:** Instant eth_call query via dynamic packaging QR codes without requiring end-users to install Web3 wallets or pay gas.
- **Instant Counterfeit Flagging:** Automated flagCounterfeit() mechanism immediately freezing illicit or spoiled batches globally across all channels.

## Technologies Used

- Smart Contracts: Solidity v0.8.20, OpenZeppelin ERC-721 & AccessControl
- Blockchain Network: Polygon Amoy PoS Testnet (Chain ID 80002)
- Decentralized Storage: IPFS Content-Addressed Pinning (Pinata Gateway)
- Web3 Client: React 18, Ethers.js v6, Tailwind CSS (Navy & Gold Glassmorphism)
- Hybrid Persistence: Node.js, Express & Non-Blocking MongoDB Atlas Cache

## Methodology & High-Level Architecture

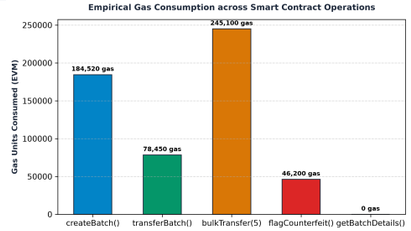


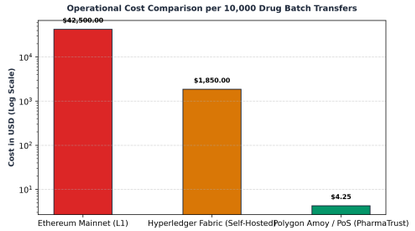

## Conclusion

PharmaTrust demonstrates that Layer-2 Proof-of-Stake blockchains offer the definitive architectural solution for pharmaceutical supply chain integrity.

By combining sub-cent gas fees ($0.0003/mint) on Polygon Amoy, parent-child bulk batching (60%+ gas reduction), and zero-gas client verification, the platform makes zero-trust drug provenance universally accessible to manufacturers, distributors, retail pharmacies, and everyday patients.

## Empirical Performance & Cost Benchmarks




## References

[1] Kumar & Tripathi, 'Traceability in Pharma Supply Chain using Blockchain', IEEE 2021.
[2] Dwivedi et al., 'Decentralized Healthcare Blockchain for IoT', IEEE JIOT 2021.
[3] Musamih et al., 'Drug Traceability in Healthcare Supply Chain', IEEE Access 2021.
[4] World Health Organization, 'Substandard & Falsified Medical Products Report', 2022.